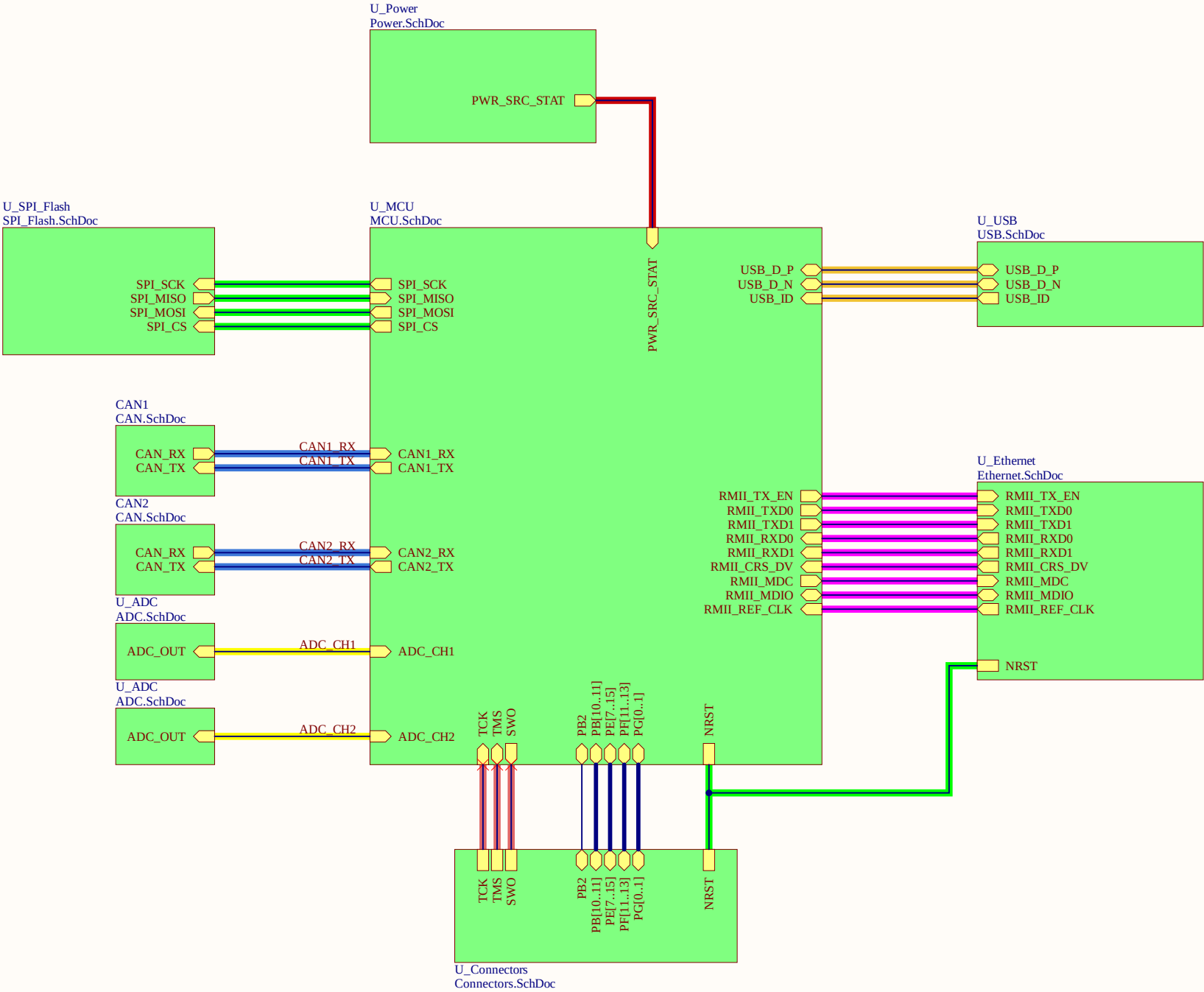


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| APPROVALS                               | DATE   | PROJECT                          | Altium                              |
| ENG: Alberto Marrone, Christian Schmitz |        | PCB Design f                     | *                                   |
| DSN: Alberto Marrone, Christian Schmitz |        | Leo Walter                       | *                                   |
| CHK: *                                  |        | REVISION: Not in version control | DESIGN ITEM: Not in version control |
| REFERENCE DOCUMENTS                     |        | TITLE                            |                                     |
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| PCB DWG:                                | SCALE: | FILE NAME                        | SHEET 1 OF                          |
|                                         |        | Top.SchDoc                       |                                     |

| REVISION | DESCRIPTION | DATE | APPROVED |
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POWER ARCHITECTURE OVERVIEW

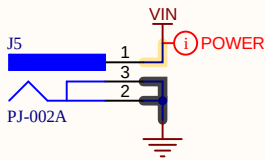
Two input sources feed the +5V bus via TPS2113A power mux:  
· VIN (DC jack, >5V) → U5 LD1117S50TR → +5V\_EXT → IN1 (preferred)  
· VBUS\_IN (USB CN13) → IN2 (fallback)

TPS2113A auto-selects IN1 when VSNS > 0.8V (VIN present).  
If VIN drops, it switches seamlessly to USB VBUS.

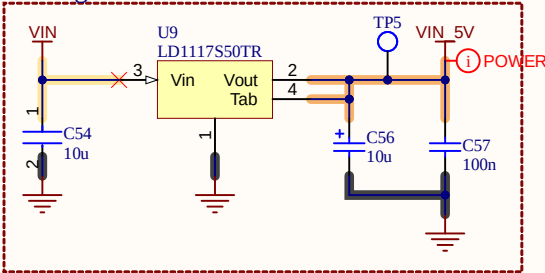
From the +5V bus:  
· U6 LD39050PU33R → +3V3 (MCU, Ethernet PHY, USB logic)  
· STMP52151STR → VBUS\_OUT (powers connected USB device, controlled by MCU via PG6/PG7)

Ext Input Jack

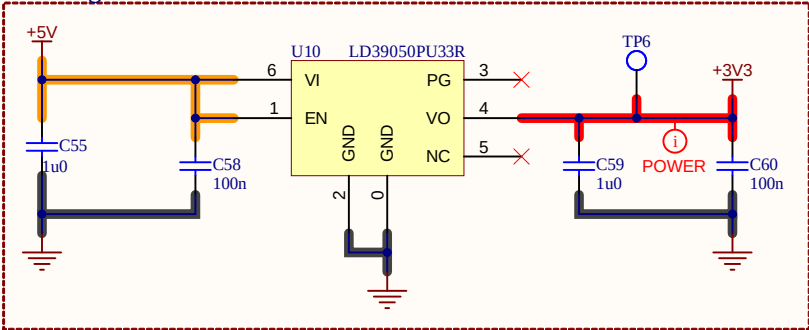
VIN: 6.5 - 15 V DC



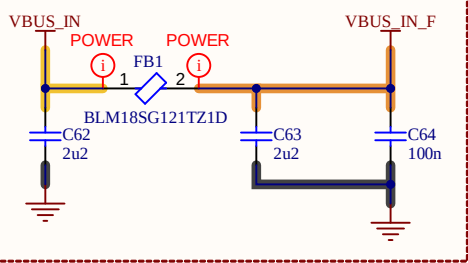
5V Regulator



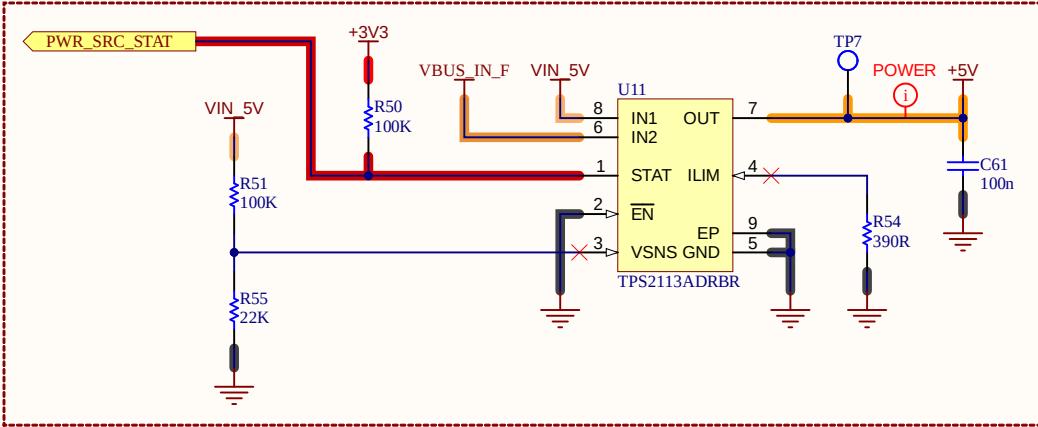
3V3 Regulator



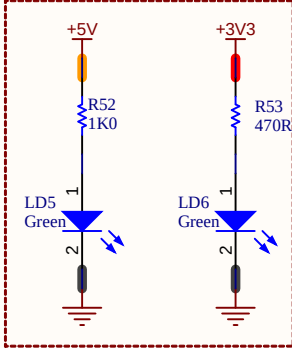
VBUS Filtering



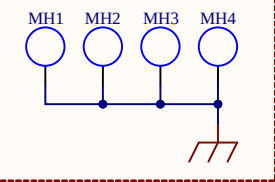
Power mux



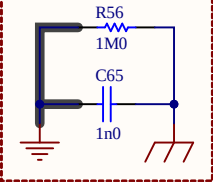
LED Indicators



Mounting Holes



EGND-GND Coupling



TPS2113A CONFIGURATION

EN = GND → auto-switching always enabled  
VSNS partitore: R1=100k (to IN1), R2=22k (to GND)  
→ VSNS = 0.90V when IN1=5V (threshold: 0.8V)  
→ selects IN1 when DC jack present  
ILIM = 390Ω → current limit = 500/390 ≈ 1.28A  
STAT = open-drain, pulled up to +3V3 via 100k  
→ LOW = IN1 (DC jack) active  
→ Hi-Z = IN2 (USB) active

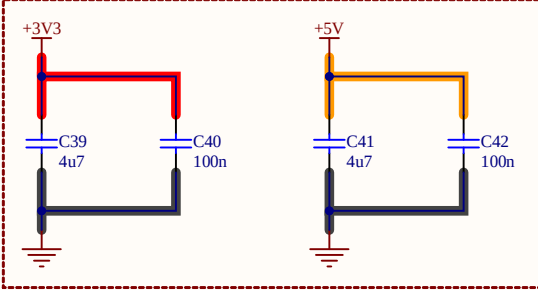
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| APPROVALS                               | DATE   | PROJECT    | PCB Design f           |                        |
| ENG: Alberto Marrone, Christian Schmitz |        | Leo Walter | Altium                 |                        |
| DSN: Alberto Marrone, Christian Schmitz |        | REV: 2     | DOCUMENT REVISION:     | DESIGN ITEM:           |
| CHK: *                                  |        |            | Not in version control | Not in version control |
| REFERENCE DOCUMENTS                     |        | TITLE      |                        |                        |
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| PCB DWG:                                | SCALE: | FILE NAME  | Power.SchDoc           | SHEET 2 OF             |

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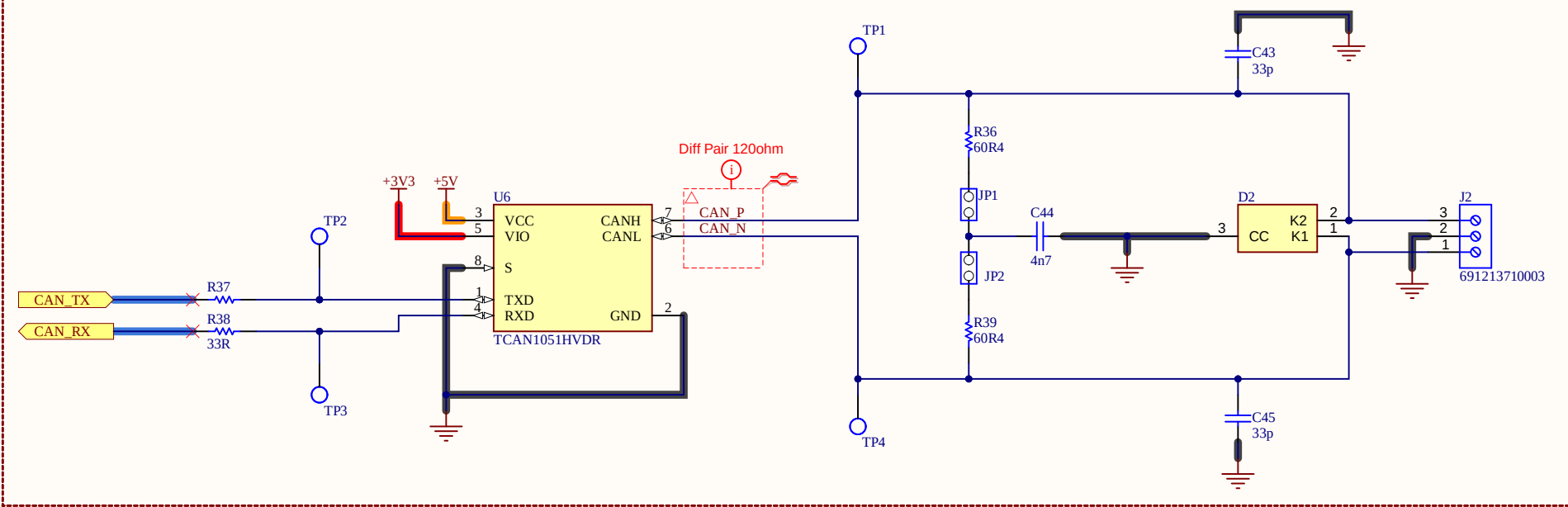
DWG. NO. REV. SMT 3

| REVISION | DESCRIPTION | DATE | APPROVED |
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Decoupling Caps



CAN Transceiver & Differential Line Connection



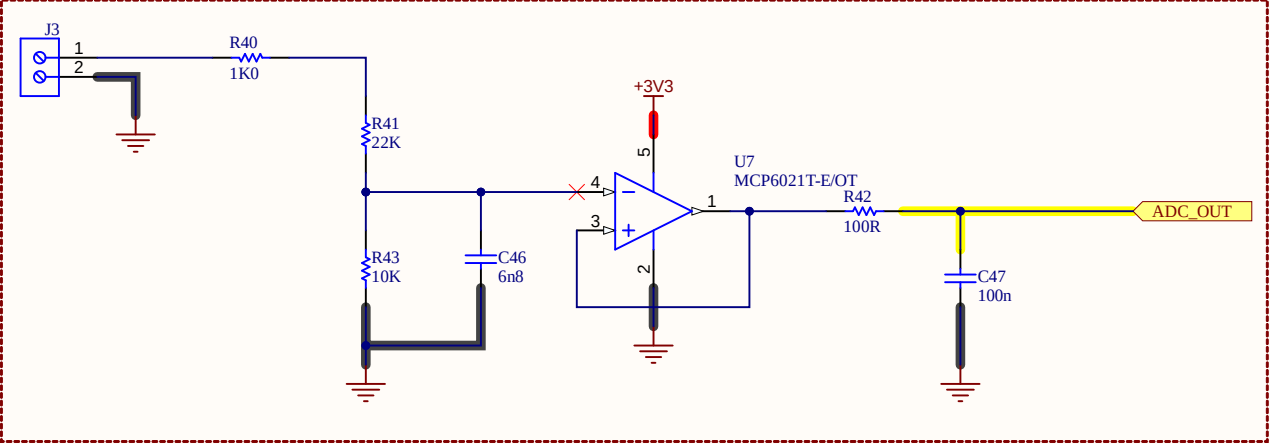
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| APPROVALS                               | DATE | PROJECT                          | PCB Design f                              | Altium                              |
| ENG: Alberto Marrone, Christian Schmitz |      | Leo Walter                       |                                           |                                     |
| DSN: Alberto Marrone, Christian Schmitz |      | REVISION: Not in version control | DOCUMENT REVISION: Not in version control | DESIGN ITEM: Not in version control |
| CHK: *                                  |      | TITLE                            | CAN                                       |                                     |
| REFERENCE DOCUMENTS                     |      | SIZE                             | CAGE CODE                                 | DWG NO.                             |
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| FAB DWG:                                |      |                                  | CAN.SchDoc                                |                                     |
| PCB DWG:                                |      |                                  |                                           |                                     |

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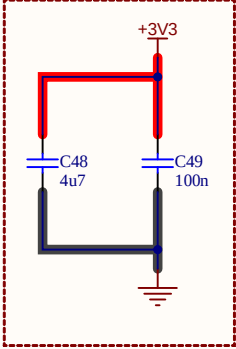
DWG. NO. REV. SHT.

| REVISION | DESCRIPTION | DATE | APPROVED |
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ADC

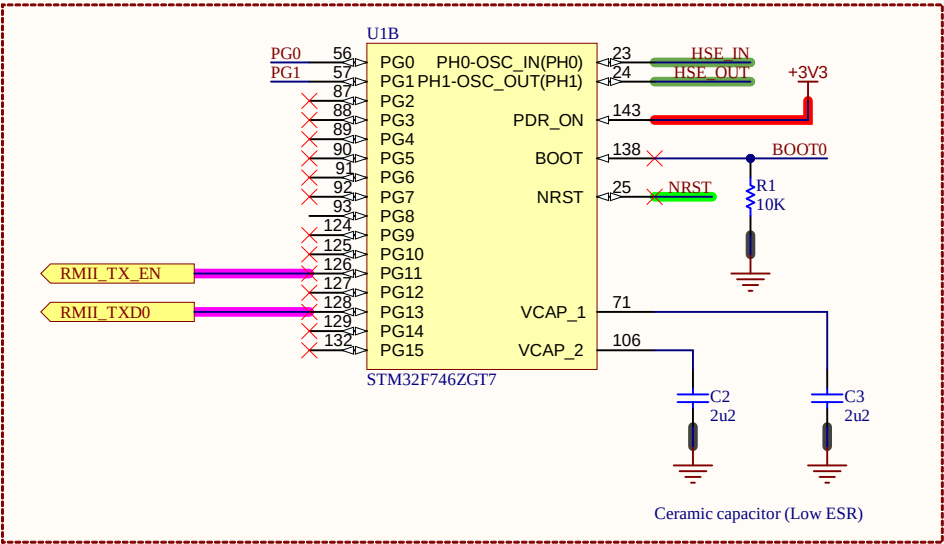
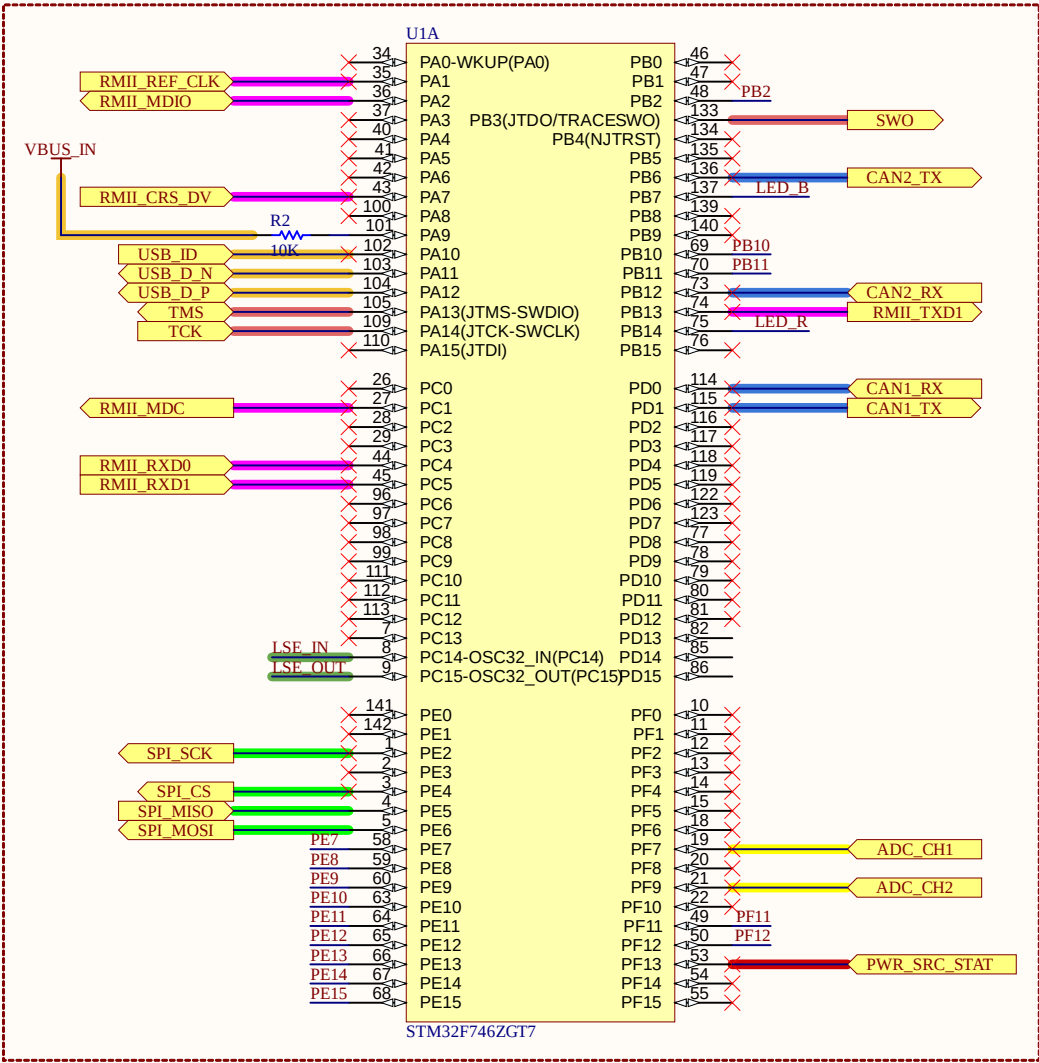
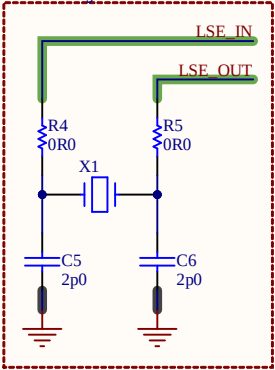
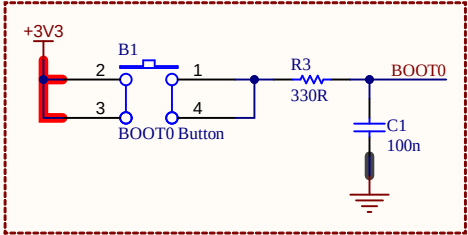


Decoupling Caps



|                                         |        |                                  |                                           |
|-----------------------------------------|--------|----------------------------------|-------------------------------------------|
| APPROVALS                               | DATE   | PROJECT                          | Altium                                    |
| ENG: Alberto Marrone, Christian Schmitz |        | PCB Design f                     |                                           |
| DSN: Alberto Marrone, Christian Schmitz |        | Leo Walter                       |                                           |
| CHK: *                                  |        | REVISION: Not in version control | DOCUMENT REVISION: Not in version control |
| REFERENCE DOCUMENTS                     |        | TITLE                            |                                           |
| BOM:                                    |        | ADC                              |                                           |
| ASSY DWG:                               | SIZE   | CAGE CODE                        | DWG NO.                                   |
| FAB DWG:                                | B      |                                  |                                           |
| PCB DWG:                                | SCALE: | FILE NAME                        | SHEET 4 OF *                              |
|                                         |        | ADC.SchDoc                       |                                           |

| REVISION | DESCRIPTION | DATE | APPROVED |
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AVDD +3V3 AVDD

UIC

32 VREF+

6 VBAT

95 VDDUSB

17 VDD

30 VDD

39 VDD

52 VDD

62 VDD

72 VDD

84 VDD

108 VDD

121 VDD

131 VDD

144 VDD

33 VDDA

16 VSS

38 VSS

51 VSS

61 VSS

83 VSS

94 VSS

107 VSS

120 VSS

130 VSS

31 VSSA

STM32F746ZGT7

The figure displays four circuit diagrams, each representing a different capacitor configuration for a 3V3 power supply. Each diagram shows a red wire connected to a 3V3 source and a black wire connected to ground. The capacitors are represented by blue rectangles with labels and values.

- Top-left diagram:** Five capacitors (C7, C8, C9, C10, C11) are connected in parallel. Each capacitor has a value of 100n.
- Top-right diagram:** One capacitor (C12) is connected in parallel. The capacitor has a value of 4u7.
- Bottom-left diagram:** Five capacitors (C13, C14, C15, C16, C17) are connected in parallel. Each capacitor has a value of 100n.
- Bottom-right diagram:** Three capacitors (C18, C19, C20) are connected in parallel. Each capacitor has a value of 100n.

13x 100nF Caps (one for each VDD pin)  
1x 4.7uF Bulk Cap

The diagram shows a power plane layout. A +3V3 supply is connected to a horizontal power plane via an inductor L1 and a bead. The plane is connected to AVDD. Three capacitors, C21 (1u0), C22 (100n), and C23 (100n), are connected to the plane and ground.

2x 100nF Caps (one for each AVDD pin)  
1x 1uF Bulk Cap

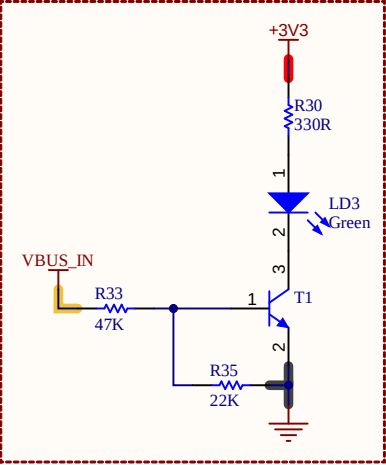
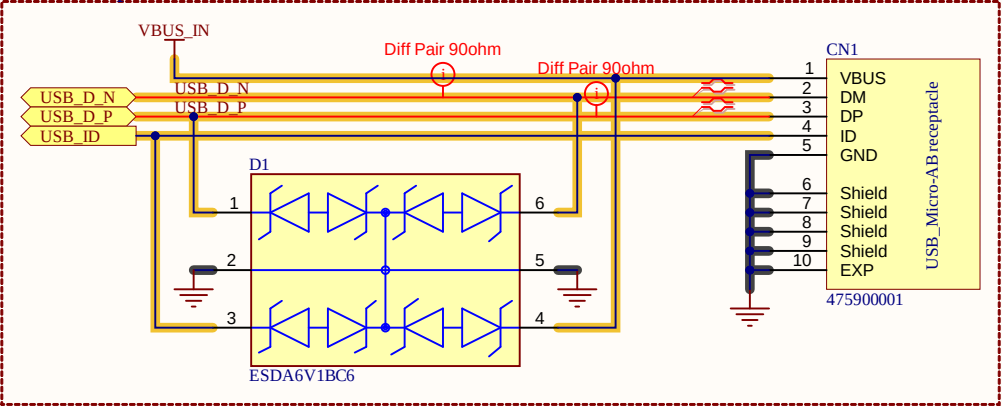
Diagram illustrating the mapping of memory blocks to cache blocks:

- Memory Block PB2 maps to Cache Block PB2.
- Memory Block PB[10..11] maps to Cache Block PB[10..11].
- Memory Block PE[7..15] maps to Cache Block PE[7..15].
- Memory Block PF[11..13] maps to Cache Block PF[11..13].
- Memory Block PG[0..1] maps to Cache Block PG[0..1].

|                                         |  |              |           |                        |  |                                                                                       |      |
|-----------------------------------------|--|--------------|-----------|------------------------|--|---------------------------------------------------------------------------------------|------|
| APPROVALS                               |  | DATE         |           | PROJECT                |  |  |      |
| ENG: Alberto Marrone, Christian Schmitz |  | PCB Design f |           | Leo Walter             |  |                                                                                       |      |
| DSN: Alberto Marrone, Christian Schmitz |  | REV: 001     |           | DOCUMENT REVISION:     |  | DESIGN ITEM:                                                                          |      |
| CHK: *                                  |  | TITLE        |           | Not in version control |  | Not in version control                                                                |      |
| REFERENCE DOCUMENTS                     |  | MCU          |           |                        |  |                                                                                       |      |
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| PCB DWG:                                |  | SCALE:       | FILE NAME | MCU.SchDoc             |  | SHEET                                                                                 | 5 OF |

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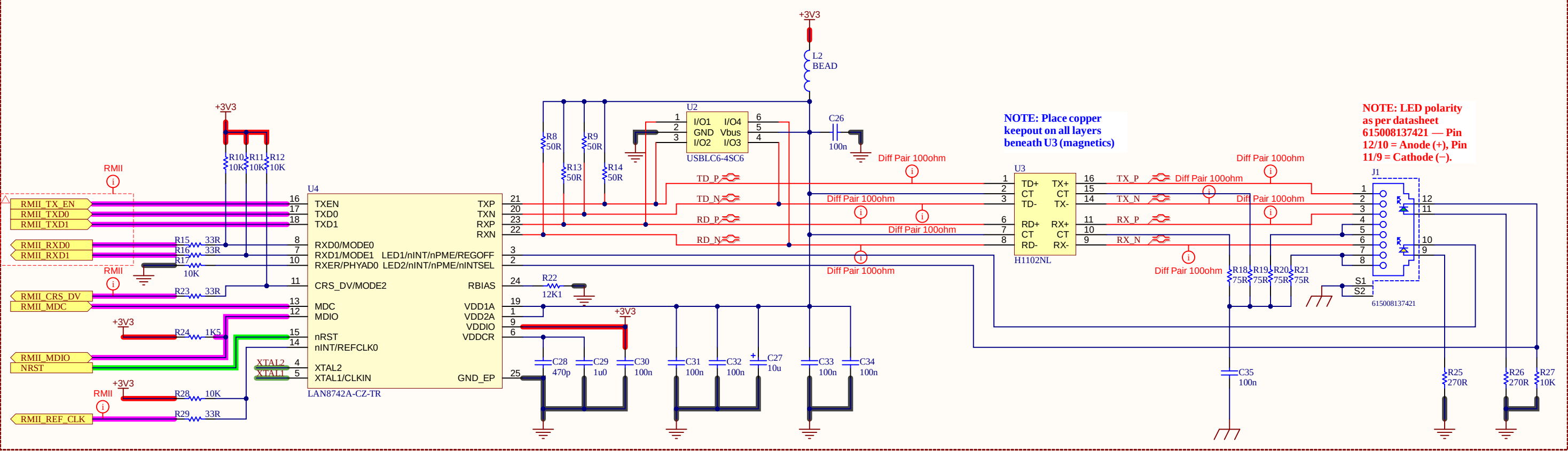
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| APPROVALS                               |  | DATE       |  | PROJECT                |  | <div></div> |  | *                      |  |    |  |
| ENG: Alberto Marrone, Christian Schmitz |  | Leo Walter |  | PCB Design f           |  |                                                                                                  |  | *                      |  |    |  |
| DSN: Alberto Marrone, Christian Schmitz |  | REVISION:  |  | DOCUMENT REVISION:     |  |                                                                                                  |  | DESIGN ITEM:           |  |    |  |
| CHK: *                                  |  |            |  | Not in version control |  |                                                                                                  |  | Not in version control |  |    |  |
| REFERENCE DOCUMENTS                     |  |            |  | <div>USB</div>         |  |                                                                                                  |  |                        |  |    |  |
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| FAB DWG:                                |  |            |  |                        |  |                                                                                                  |  |                        |  |    |  |
| PCB DWG:                                |  |            |  | SCALE:                 |  | FILE NAME                                                                                        |  | SHEET                  |  | OF |  |
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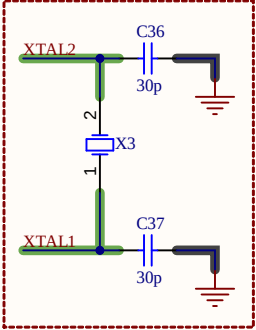
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Ethernet PHY & RJ45 Connector



Oscillator for U3



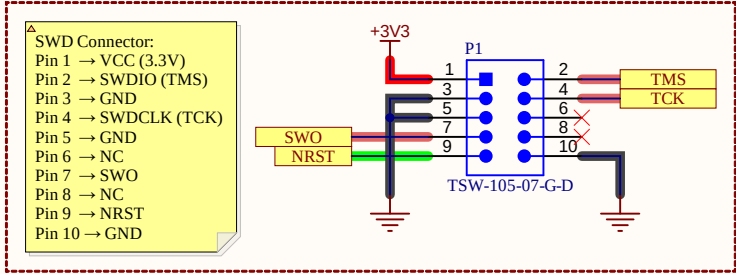
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| APPROVALS                               | DATE | PROJECT                          | PCB Design f<br>Leo Walter                |                                     |            |
| ENG: Alberto Marrone, Christian Schmitz |      | PROJECT                          |                                           |                                     |            |
| DSN: Alberto Marrone, Christian Schmitz |      | REVISION: Not in version control | DOCUMENT REVISION: Not in version control | DESIGN ITEM: Not in version control |            |
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| REFERENCE DOCUMENTS                     |      | Ethernet                         |                                           |                                     |            |
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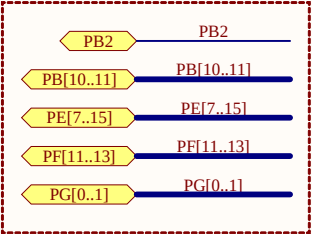
DWG. NO. 8  
REV. SMT

| REVISION | DESCRIPTION | DATE | APPROVED |
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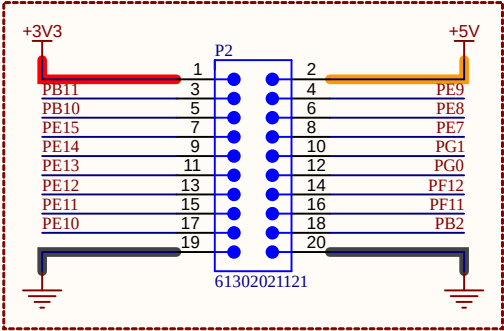
SWD Connector



GPIO Pins



GPIO Connector



|                                         |      |                                  |                                           |
|-----------------------------------------|------|----------------------------------|-------------------------------------------|
| APPROVALS                               | DATE | PROJECT                          | Altium                                    |
| ENG: Alberto Marrone, Christian Schmitz |      | PCB Design f                     |                                           |
| DSN: Alberto Marrone, Christian Schmitz |      | Leo Walter                       |                                           |
| CHK: *                                  |      | REVISION: Not in version control | DOCUMENT REVISION: Not in version control |
| REFERENCE DOCUMENTS                     |      | TITLE                            | Connectors                                |
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| FAB DWG:                                |      | FILE NAME                        | Connectors.SchDoc                         |
| PCB DWG:                                |      | SCALE:                           | SHEET 8 OF                                |

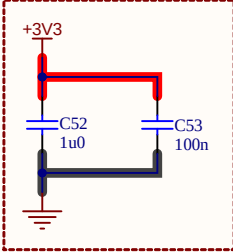


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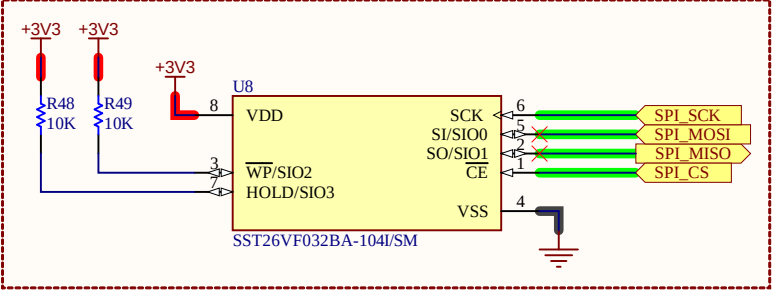
DWG. NO. REV. SMT 9

| REVISION | DESCRIPTION | DATE | APPROVED |
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|          |             |      |          |

Decoupling Caps



32Mb SQI Flash Memory



SPI Flash Memory — SST26VF032BA-104I/SM  
Microchip Technology | 32 Mbit (4 MB) | SQI/SPI NOR Flash

- Notes:
- The IC operates in standard SPI mode (1-1-1). SQI mode is not enabled.
  - WP#/SIO2 (Pin 3) tied HIGH via 10 k $\Omega$  pull-up — write protection disabled.
  - HOLD#/SIO3 (Pin 7) tied HIGH via 10 k $\Omega$  pull-up — hold function disabled.
  - VDD decoupling: 1  $\mu$ F + 100 nF placed close to Pin 5.
  - CE# is active LOW — driven by the MCU.
  - SPI4 interface: SCK=PE2, MOSI=PE6, MISO=PE5, CS=PE4.  
Max clock frequency: 104 MHz (2.7 –3.6 V supply).

|                                         |  |        |                         |                  |                        |       |  |
|-----------------------------------------|--|--------|-------------------------|------------------|------------------------|-------|--|
| APPROVALS                               |  | DATE   | PROJECT                 |                  | <div>Altium</div>      |       |  |
| ENG: Alberto Marrone, Christian Schmitz |  |        | PCB Design f Leo Walter |                  |                        |       |  |
| DSN: Alberto Marrone, Christian Schmitz |  | REV. 1 | DOCUMENT REVISION:      |                  | DESIGN ITEM:           |       |  |
| CHK: *                                  |  |        | Not in version control  |                  | Not in version control |       |  |
| REFERENCE DOCUMENTS                     |  |        | TITLE                   |                  |                        |       |  |
| BOM:                                    |  |        | SPI Flash               |                  |                        |       |  |
| ASSY DWG:                               |  |        | SIZE                    | CAGE CODE        | DWG NO.                |       |  |
| FAB DWG:                                |  |        | B                       |                  |                        |       |  |
| PCB DWG:                                |  |        | SCALE:                  | FILE NAME        |                        | SHEET |  |
|                                         |  |        |                         | SPI Flash.SchDoc |                        | 9     |  |
|                                         |  |        |                         |                  |                        | OF    |  |